

Let $\{(x_i^{(s)})_{i=1}^{n_x}, (y_j^{(s)})_{j=1}^{n_y}\}_{s=1}^S$ be configurations and let T_k denote Chebyshev polynomials. Define ACE pooled features

$$a_x^{(s)}(k) = \sum_{i=1}^{n_x} T_k(x_i^{(s)}), \quad k = 0, \dots, d_x, \quad a_y^{(s)}(\ell) = \sum_{j=1}^{n_y} T_\ell(y_j^{(s)}), \quad \ell = 0, \dots, d_y.$$

Given targets $E^{(s)} \in \mathbb{R}$, fit a bilinear model with Tucker-structured coefficient matrix

$$E^{(s)} \approx (a_x^{(s)})^\top C a_y^{(s)}, \quad C \approx AGB^\top,$$

where $A \in \mathbb{R}^{(d_x+1) \times R_x}$, $B \in \mathbb{R}^{(d_y+1) \times R_y}$, and $G \in \mathbb{R}^{R_x \times R_y}$.

Define reduced features $u^{(s)} = (a_x^{(s)})^\top A \in \mathbb{R}^{R_x}$ and $v^{(s)} = (a_y^{(s)})^\top B \in \mathbb{R}^{R_y}$, so that

$$E^{(s)} \approx (u^{(s)})^\top G v^{(s)}.$$

ALS updates alternate between two least-squares steps followed by truncation via SVD. Fix B and solve for $\tilde{G} \in \mathbb{R}^{(d_x+1) \times R_y}$ from

$$\tilde{G} \in \arg \min_{Z \in \mathbb{R}^{(d_x+1) \times R_y}} \sum_{s=1}^S \left(E^{(s)} - (a_x^{(s)})^\top Z v^{(s)} \right)^2, \quad v^{(s)} = (a_y^{(s)})^\top B.$$

Compute the truncated SVD $\tilde{G} \approx U \Sigma V^\top$ with rank R_x and set

$$A \leftarrow U_{(:,1:R_x)}, \quad G \leftarrow \Sigma_{1:R_x,1:R_x} V_{(:,1:R_x)}^\top.$$

Fix A and solve for $\tilde{H} \in \mathbb{R}^{(d_y+1) \times R_x}$ from

$$\tilde{H} \in \arg \min_{Z \in \mathbb{R}^{(d_y+1) \times R_x}} \sum_{s=1}^S \left(E^{(s)} - (a_y^{(s)})^\top Z u^{(s)} \right)^2, \quad u^{(s)} = (a_x^{(s)})^\top A.$$

Compute the truncated SVD $\tilde{H} \approx \hat{U} \hat{\Sigma} \hat{V}^\top$ with rank R_y and set

$$B \leftarrow \hat{U}_{(:,1:R_y)}, \quad G \leftarrow \hat{V}_{(:,1:R_y)} \hat{\Sigma}_{1:R_y,1:R_y}.$$

After convergence, the learned coefficient matrix is $C = AGB^\top$ and predictions are

$$\hat{E}^{(s)} = (a_x^{(s)})^\top A G B^\top a_y^{(s)}.$$